1 End Behaviour of Functions

Horizontal and Slant Asymptotes

Warm-up. Find the critical numbers of $p(x) = x^3 - 3x^2 + 2$. Classify each using the first and second derivative tests.

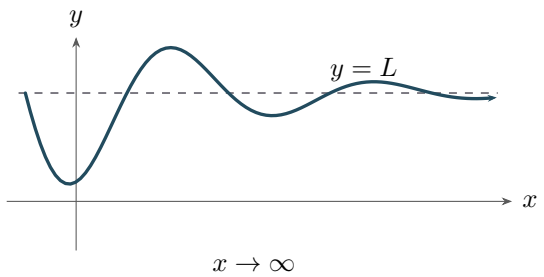
$p'(x) = 3x(x-2)$ has signs $+, -, +$ across $0, 2$. At 0, $p''(0) = -6$: local max. $(0, 2)$. At 2, $p''(2) = 6$: local min. $(2, -2)$.

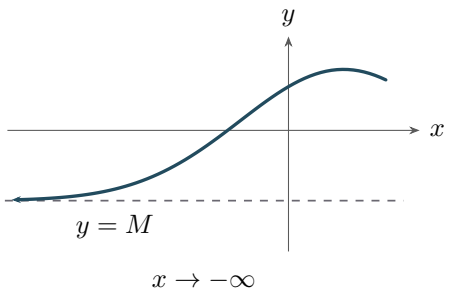
Definition (Limit at infinity). We write

$$\lim_{x\to\infty} f(x) = L \quad \text{if } f(x) \to L \text{ as } x \to \infty.$$

Similarly, we write $\lim_{x\to-\infty} f(x) = L$ if $f(x) \to L$ as $x \to -\infty$.

Definition (Horizontal asymptote). The line $y = L$ is a horizontal asymptote of f if $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$.





Find the limits at both ends and identify any horizontal asymptotes.

Example 1. $f(x) = \dfrac{x^2-1}{x} = x - \dfrac{1}{x}$.

$$\lim_{x\to-\infty} f(x) = -\infty, \qquad \lim_{x\to\infty} f(x) = +\infty. \qquad \text{No horizontal asymptote.}$$

Example 2. $f(x) = \dfrac{x^2-4}{x^2-x+1}$.

$$\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \frac{1 - 4/x^2}{1 - 1/x + 1/x^2} = 1. \qquad \text{Horizontal asymptote: } y = 1.$$

Example 3. $f(x) = \dfrac{x+1}{x^3-1}$.

$$\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \frac{1/x^2 + 1/x^3}{1 - 1/x^3} = 0. \qquad \text{Horizontal asymptote: } y = 0.$$

Theorem (End behaviour of a rational function). Let

$$p(x) = a_m x^m + \cdots + a_0, \qquad q(x) = b_n x^n + \cdots + b_0, \qquad a_m b_n \neq 0.$$

At either end, p/q has the same limit as the quotient of its leading terms:

$$\lim_{x \to \pm\infty} \frac{p(x)}{q(x)} = \lim_{x \to \pm\infty} \frac{a_m}{b_n}\, x^{m-n}.$$

If $m < n$, both limits are 0. If $m = n$, both are a_m/b_n. If $m > n$, the sign of $(a_m/b_n)x^{m-n}$ determines $+\infty$ or $-\infty$ at each end.

Proof. Divide out the leading powers:

$$\frac{p(x)}{q(x)} = x^{m-n} \frac{a_m + a_{m-1}/x + \cdots + a_0/x^m}{b_n + b_{n-1}/x + \cdots + b_0/x^n}.$$

The fraction on the right tends to a_m/b_n at both ends. □

Example 4. Find the limits as $x \to +\infty$ and as $x \to -\infty$ for each function.

(a) $f(x) = \dfrac{1 - x^3}{2x^3 + x - 1}$. Dividing the numerator and denominator by x^3 gives $\frac{1/x^3 - 1}{2 + 1/x^2 - 1/x^3}$. At either end, the numerator tends to -1 and the denominator tends to 2. Thus

$$\lim_{x \to +\infty} f(x) = -\frac{1}{2}, \qquad \lim_{x \to -\infty} f(x) = -\frac{1}{2}.$$

(b) $g(x) = \dfrac{x^{2026} + x^{2025} + \cdots + x + 1}{2026x^{2026} - 2025x^{2025} + \cdots - x + 1}$. Both polynomials have degree 2026. Dividing by x^{2026} makes every lower-degree term tend to zero at either end. The leading coefficients give

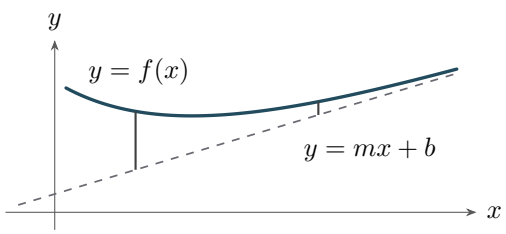
$$\lim_{x \to +\infty} g(x) = \frac{1}{2026}, \qquad \lim_{x \to -\infty} g(x) = \frac{1}{2026}.$$

(c) $h(x) = \dfrac{x^{2027} + x^{2026} + \cdots + x + 1}{2026x^{2026} - 2025x^{2025} + \cdots - x + 1}$. Here the numerator has one more power of x than the denominator. The quotient of the leading terms is $\frac{x^{2027}}{2026x^{2026}} = \frac{x}{2026}$. On the right, $x/2026$ grows without bound. On the left, it is negative and its magnitude grows without bound. Hence

$$\lim_{x \to +\infty} h(x) = +\infty, \qquad \lim_{x \to -\infty} h(x) = -\infty.$$

Definition (Slant asymptote). A line $y = mx + b$, with $m \neq 0$, is a slant asymptote of f if

$$\lim_{x \to \infty} \big(f(x) - (mx + b)\big) = 0 \quad \text{or} \quad \lim_{x \to -\infty} \big(f(x) - (mx + b)\big) = 0.$$



Vertical gap
$\big|f(x) - (mx + b)\big| \to 0.$

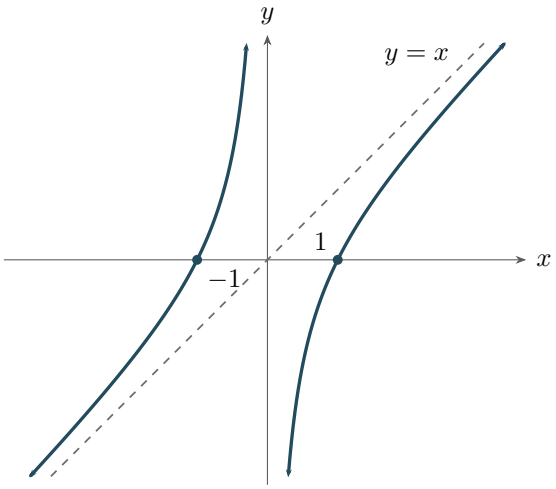
Example 5. Find the slant asymptote of $f(x) = \dfrac{x^2 - 1}{x}$.

$$f(x) = x - \frac{1}{x}, \qquad f(x) - x = -\frac{1}{x}.$$

Thus $\lim\limits_{x \to \pm\infty} (f(x) - x) = 0$. At $x = 0$,

$$\lim_{x \to 0^-} f(x) = +\infty, \qquad \lim_{x \to 0^+} f(x) = -\infty.$$

The left branch lies above $y = x$, and the right branch lies below it.



Example 6. Find the slant asymptote of $g(x) = \dfrac{x^3}{x^2 + 1}$. *Hint.* $x(x^2 + 1) = x^3 + x$.

$g(x) - x = -x/(x^2 + 1) \to 0$ at both ends. The slant asymptote is $y = x$.

Theorem (Slant asymptote by division). If $\deg P = \deg Q + 1$, the line given by the quotient in polynomial division is a slant asymptote of P/Q at both ends.

Proof. Write $P = (mx + b)Q + R$ with $R = 0$ or $\deg R < \deg Q$. Then $\dfrac{P(x)}{Q(x)} - (mx + b) = \dfrac{R(x)}{Q(x)} \to 0$ as $x \to \pm\infty$. □

Example 7. Find both end limits and all horizontal asymptotes.

$$\text{(a) } \frac{\sqrt{4x^2 + 1}}{x} \qquad \text{(b) } \frac{2x^2 + 4}{\sqrt{x^4 + 1}} \qquad \text{(c) } \frac{e^{2x} - 1}{e^{x^2} + x + 1}.$$

(a) $\frac{|x|}{x}\sqrt{4 + x^{-2}} \to -2$ at the left and 2 at the right: $y = -2,\ y = 2$.

(b) $(2 + 4/x^2)/\sqrt{1 + x^{-4}} \to 2$ at both ends: $y = 2$.

(c) Divide by e^{x^2}. The numerator tends to 0 since $2x - x^2 = 1 - (x - 1)^2$; the denominator tends to 1. Both limits are 0: $y = 0$.

2 Curve Sketching

We now have the tools to sketch a graph: limits tell us where its branches go, and derivatives tell us how they rise, fall, and bend.

Curve Sketching Steps

1. **Domain.** Identify the x-values where $f(x)$ is defined.
2. **Intercepts.** Solve $f(x) = 0$. Find $f(0)$ if it exists.
3. **Symmetry.** Check whether f is even, odd, or periodic.
4. **Asymptotes.** Check one-sided limits at domain breaks and limits at both ends.
5. **Increase and decrease.** Find the intervals where $f' > 0$ or $f' < 0$.
6. **Local extrema.** Classify critical numbers with a derivative test.
7. **Concavity and inflection points.** Check the sign of f''. Mark points where concavity changes.
8. **Sketch.** Draw asymptotes, plot key points, and trace each branch.

Example 8. Sketch the graph of $f(x) = \dfrac{2x^2}{x^2 - 1}$.

1. Domain.

$$(-\infty, -1) \cup (-1, 1) \cup (1, \infty) \quad \text{also written} \quad \mathbb{R} \setminus \{-1, 1\}.$$

2. Intercepts. Both axes meet the graph at $(0, 0)$.

3. Symmetry. $f(-x) = f(x)$. The graph is symmetric about the y-axis.

4. Asymptotes. Factor $x^2 - 1 = (x-1)(x+1)$. The numerator approaches 2 at both excluded inputs, so the one-sided limits are

$$\lim_{x \to -1^-} f(x) = +\infty, \qquad \lim_{x \to -1^+} f(x) = -\infty,$$

$$\lim_{x \to 1^-} f(x) = -\infty, \qquad \lim_{x \to 1^+} f(x) = +\infty.$$

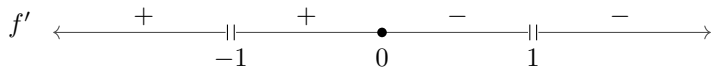
Thus $x = -1$ and $x = 1$ are vertical asymptotes. At both ends,

$$\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \left(2 + \frac{2}{x^2 - 1}\right) = 2.$$

The graph approaches $y = 2$ from above.

Example 8 continued $f(x) = \dfrac{2x^2}{x^2-1}$

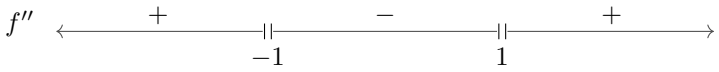
5. Increase and decrease. $f'(x) = \dfrac{-4x}{(x^2-1)^2}$.



Increasing on each interval in	$(-\infty,-1)\cup(-1,0)$,	also written $\{x \in \mathbb{R} : x < 0,\ x \neq -1\}$.
Decreasing on each interval in	$(0,1)\cup(1,\infty)$,	also written $\{x \in \mathbb{R} : x > 0,\ x \neq 1\}$.

6. Local extrema. At $x = 0$, f' changes from $+$ to $-$. The only local extremum is a maximum at $(0,0)$.

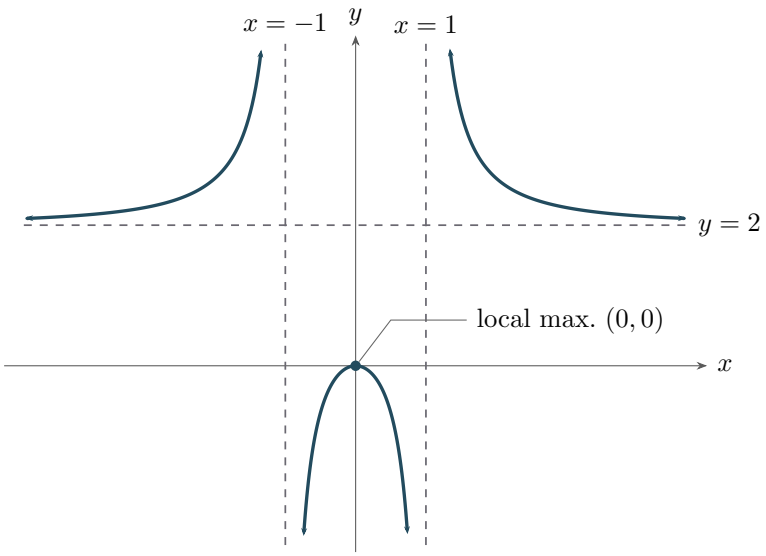
7. Concavity and inflection points. $f''(x) = \dfrac{4(3x^2+1)}{(x^2-1)^3}$.



Concave up	$(-\infty,-1)\cup(1,\infty)$,	also written $\{x \in \mathbb{R} : \lvert x\rvert > 1\}$.
Concave down	$(-1,1)$,	also written $\{x \in \mathbb{R} : \lvert x\rvert < 1\}$.

Both changes of concavity occur at excluded inputs. There are no inflection points.

8. Sketch.



Example 9. Sketch the graph of $g(x) = \dfrac{x^3}{x^2+1}$.

1. Domain. $(-\infty, \infty)$, also written $\mathbb{R}$.

2. Intercepts. Both intercepts are $(0, 0)$.

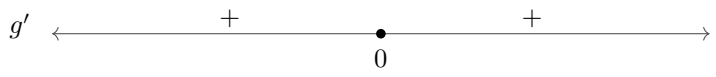
3. Symmetry. $g(-x) = -g(x)$. The graph is symmetric about the origin.

4. Asymptotes. The denominator $x^2 + 1$ is positive everywhere. At both ends,

$$\lim_{x\to\pm\infty} \big(g(x) - x\big) = \lim_{x\to\pm\infty} \frac{-x}{x^2+1} = 0.$$

The slant asymptote is $y = x$. The graph lies above it for $x < 0$ and below it for $x > 0$.

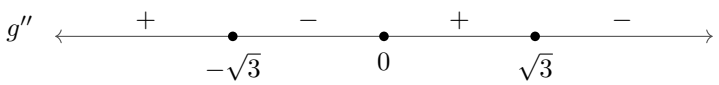
5. Increase and decrease. $g'(x) = \dfrac{x^2(x^2+3)}{(x^2+1)^2}$.



The function is strictly increasing on $(-\infty, \infty)$, also written $\mathbb{R}$.

6. Local extrema. The tangent at the origin is horizontal. The derivative stays positive on either side, so there are no local extrema.

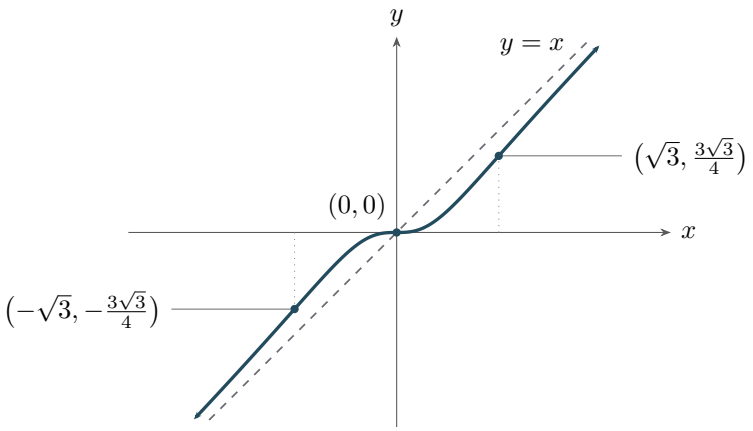
7. Concavity and inflection points. $g''(x) = \dfrac{2x(3-x^2)}{(x^2+1)^3}$.



Concave up on $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$, also written $\{x \in \mathbb{R} : x < -\sqrt{3} \text{ or } 0 < x < \sqrt{3}\}$.

Concave down on $(-\sqrt{3}, 0) \cup (\sqrt{3}, \infty)$, also written $\{x \in \mathbb{R} : -\sqrt{3} < x < 0 \text{ or } x > \sqrt{3}\}$.

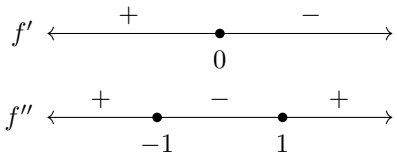
8. Sketch. The three marked points are inflection points.

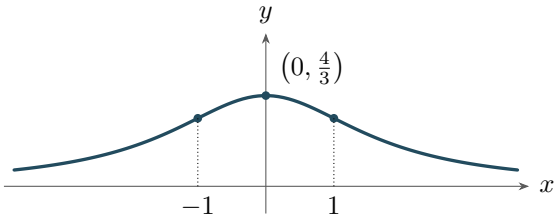


Curve Sketching Practice

1. $f(x) = \dfrac{4}{x^2+3}$, $\quad f'(x) = \dfrac{-8x}{(x^2+3)^2}$, $\quad f''(x) = \dfrac{24(x^2-1)}{(x^2+3)^3}$.

Domain $(-\infty, \infty) = \mathbb{R}$. Even. The only intercept is $(0, 4/3)$. Since $\lim\limits_{x\to\pm\infty} 4/(x^2+3) = 0$, both ends approach $y = 0$ from above.





Increasing on $(-\infty, 0)$, decreasing on $(0, \infty)$. Absolute maximum at $(0, 4/3)$. Concave up on $(-\infty, -1)$ and $(1, \infty)$, down on $(-1, 1)$. Inflection points $(-1, 1)$ and $(1, 1)$.

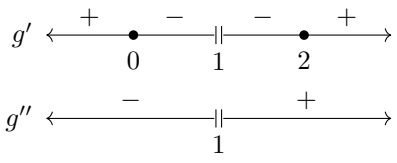
2. $g(x) = \dfrac{x^2}{x-1} = x + 1 + \dfrac{1}{x-1}$, $\quad g'(x) = \dfrac{x(x-2)}{(x-1)^2}$, $\quad g''(x) = \dfrac{2}{(x-1)^3}$.

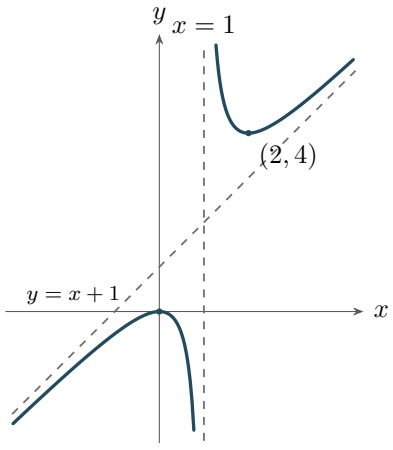
Domain $(-\infty, 1) \cup (1, \infty) = \mathbb{R} \setminus \{1\}$. Both intercepts are $(0, 0)$. The graph has half-turn symmetry about $(1, 2)$.

At $x = 1$, $\lim\limits_{x\to 1^-} g(x) = -\infty$ and $\lim\limits_{x\to 1^+} g(x) = +\infty$. For the slant asymptote,

$$\lim_{x\to\pm\infty} \big[g(x) - (x+1)\big] = \lim_{x\to\pm\infty} \frac{1}{x-1} = 0.$$

The branch for $x < 1$ lies below $y = x + 1$, and the branch for $x > 1$ lies above it.





Increasing on $(-\infty, 0)$ and $(2, \infty)$, decreasing on $(0, 1)$ and $(1, 2)$. Local maximum $(0, 0)$, local minimum $(2, 4)$. Concave down on $(-\infty, 1)$, up on $(1, \infty)$. No inflection points.